

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- AI and Data Engineering professional with **5+ years of experience** designing and delivering enterprise data platforms, Generative AI solutions, machine learning workflows, cloud-native services, and scalable data pipelines across insurance, healthcare, and technology environments.
- Hands-on experience developing and validating **Generative AI agents, LLM-powered applications, RAG solutions, prompt-driven workflows, and enterprise AI integrations**, with emphasis on reliability, scalability, governance, and measurable business outcomes.
- Strong engineering experience with **Python, SQL, PySpark, Palantir Foundry/AIP, Power Automate, REST APIs, microservices, ETL/ELT, data modeling, and cloud platforms**, transforming structured and unstructured data into actionable insights.
- Experienced implementing **LLM evaluation frameworks, defined evaluation metrics, prompt engineering, AI governance controls, data-quality frameworks, observability, and automated validation pipelines** for enterprise AI solutions.
- Collaborated with **data scientists, ML engineers, product teams, architects, and business stakeholders** to analyze complex problems, interpret data, translate requirements into technical solutions, and provide actionable recommendations.
- Experienced integrating AI and data solutions with **CI/CD pipelines, Power Automate, enterprise APIs, workflow platforms, cloud services, and governed data sources** while maintaining documentation, security, privacy, and professional technical standards.
- Demonstrated technical leadership through **mentoring junior engineers, design reviews, knowledge sharing, documentation, critical thinking, stakeholder communication, and multidisciplinary collaboration** in fast-moving and ambiguous delivery environments.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & AI Engineering:** Python, SQL, PySpark, JavaScript, TypeScript, Java, C#, FastAPI, Flask, REST APIs, Microservices, Object-Oriented Programming, Distributed Systems.
- **Generative AI & LLMs:** Generative AI, LLMs, AI Agents, Agentic Workflows, RAG, Prompt Engineering, OpenAI, Anthropic Claude, Google Gemini, LangChain, LlamaIndex, Embeddings, Vector Databases, Semantic Search.
- **AI Evaluation & Governance:** LLM Evaluation Frameworks, Evaluation Metrics, Prompt Evaluation, Response Quality, Hallucination Analysis, AI Governance Controls, Responsible AI, Data Quality Frameworks, Model Monitoring, Observability.
- **Data Engineering & Analytics:** Palantir Foundry/AIP, ETL/ELT, Data Pipelines, Data Modeling, Structured/Unstructured Data, CDC, Data Reconciliation, PostgreSQL, MySQL, SQL Server, Oracle, Data Visualization, Statistical Analysis Concepts.
- **Automation & Integration:** Microsoft Power Automate, Workflow Automation, REST APIs, API Integration, Event-Driven Architecture, OAuth2/OIDC, JWT, Enterprise Data Patterns, System Integration, Data Validation, Business Process Automation.
- **Cloud & DevOps:** AWS, Azure, GCP Concepts, AWS Lambda, S3, IAM, Docker, Terraform, Git, Azure DevOps, Jenkins, TeamCity, GitLab CI/CD, Continuous Integration, Continuous Delivery.
- **Leadership & Delivery:** Technical Mentoring, Critical Thinking, Complex Problem Analysis, Actionable Insights, Client Collaboration, Multidisciplinary Teams, Technical Documentation, Professional Standards, Agile/Scrum, Solution Design.

EXPERIENCE

Software Engineer AI PCMI LLC (Full Time)

Oct 2024 – Present
Dallas, TX, USA

- Designed and delivered enterprise **Generative AI agents and data solutions** using Palantir Foundry/AIP, Python, FastAPI, SQL, LLMs, RAG, and cloud services to automate complex insurance workflows and transform operational data into actionable business insights.
- Built and maintained scalable **structured and unstructured data pipelines** using Python, PySpark, SQL, ETL/ELT, CDC, data modeling, and Palantir transformations, integrating transactional systems, APIs, and governed enterprise data sources.
- Implemented **LLM evaluation frameworks and AI governance controls** using defined metrics for response quality, groundedness, relevance, latency, prompt effectiveness, and model behavior, improving reliability and consistency of production-oriented AI agents.
- Developed agentic and RAG workflows incorporating **prompt engineering, embeddings, vector databases, semantic retrieval, enterprise APIs, tool integrations, and human-in-the-loop governance** for data-intensive insurance use cases.
- Integrated AI workflows with **Microsoft Power Automate, REST APIs, microservices, CI/CD pipelines, cloud services, and enterprise applications**, enabling reliable workflow automation and controlled movement of business data across systems.
- Analyzed complex data and AI issues with product owners, data engineers, architects, and business stakeholders, using **critical thinking, data interpretation, root-cause analysis, and business context** to produce actionable recommendations and scalable technical solutions.
- Provided technical leadership through **design reviews, mentoring, documentation, reusable engineering patterns, governance guidance, and knowledge sharing**, reinforcing high standards for reliability, scalability, security, data quality, and professional engineering practices.

Environments: Palantir Foundry, Palantir AIP, Python, FastAPI, PySpark, SQL, Generative AI, LLMs, RAG, Prompt Engineering, LLM Evaluation Frameworks, AI Governance, Power Automate, REST APIs, Vector Databases, ETL/ELT, CDC, AWS, Azure, Docker, Terraform, CI/CD, Data Quality, Observability.

Software Engineer AI Client: CVS

Dec 2023 – Sept 2024
Remote, USA

- Designed and developed healthcare AI and data applications using **Python, FastAPI, AWS, SQL, REST APIs, and microservices**, supporting scalable processing of regulated healthcare information and integration with enterprise business workflows.
- Developed **LLM-powered agents, RAG applications, prompt-driven workflows, embeddings, and semantic retrieval** to extract information from structured and unstructured healthcare content and deliver contextual insights to downstream applications.
- Built enterprise data pipelines using **Python, SQL, cloud services, APIs, data transformations, and relational data models**, improving accessibility and consistency of information consumed by analytics and AI-enabled applications.
- Implemented **LLM evaluation and governance processes** using defined quality metrics, prompt evaluation, groundedness analysis, response consistency, monitoring, and documentation to support reliable AI-enabled healthcare solutions.
- Integrated cloud applications with **AWS Lambda, Power Automate concepts, enterprise APIs, OAuth2/OIDC, JWT, workflow services, and CI/CD pipelines**, addressing security, privacy, data handling, and scalable deployment requirements.
- Collaborated with **data scientists, ML engineers, architects, product teams, and business stakeholders** to analyze complex requirements, interpret operational data, evaluate solution alternatives, and translate findings into actionable technical recommendations.
- Created reusable **Python libraries, data integration components, prompt patterns, technical documentation, governance artifacts, and implementation guides** while sharing engineering practices and supporting junior team members across distributed delivery teams.

Environments: Python, FastAPI, AWS, AWS Lambda, SQL, Generative AI, LLMs, RAG, Prompt Engineering, LangChain, Vector Databases, REST APIs, Microservices, Power Automate Concepts, OAuth2/OIDC, JWT, CI/CD, LLM Evaluation, AI Governance, Data Quality, Cloud Monitoring.

Software Engineer Infor

Sept 2021 – Dec 2022
Hyderabad, India

- Designed and developed enterprise data applications using **Python, FastAPI, Flask, SQL, REST APIs, and microservices**, transforming operational datasets into scalable application services and actionable information for business users.
- Engineered robust **ETL/ELT and data-processing pipelines** using Python, Pandas, PySpark, SQL, PostgreSQL, MySQL, and Oracle to ingest, transform, reconcile, and model large structured datasets across enterprise platforms.
- Developed reusable backend and workflow services for **data transformation, API orchestration, asynchronous processing, business rules, enterprise integrations, and automated workflows**, improving scalability and maintainability across applications.
- Applied **data-quality frameworks, validation rules, reconciliation techniques, metadata checks, and monitoring** to maintain accuracy, consistency, completeness, and reliability across upstream and downstream data flows.
- Integrated internal platforms through **REST APIs, event-driven patterns, authentication, workflow services, CI/CD, and cloud-native deployment approaches**, enabling reliable exchange of business data across distributed applications.
- Partnered with product managers, data engineers, architects, and business stakeholders to **analyze complex problems, interpret data trends, assess technical trade-offs, and provide actionable insights** aligned with broader project and business objectives.
- Contributed to technical excellence through **architecture discussions, reusable libraries, documentation, code reviews, knowledge sharing, and mentoring**, promoting consistent engineering standards across multidisciplinary teams.

Environments: Python, FastAPI, Flask, Pandas, PySpark, SQL, PostgreSQL, MySQL, Oracle, ETL/ELT, Data Pipelines, Data Modeling, REST APIs, Microservices, Data Quality Frameworks, Event-Driven Architecture, Docker, Jenkins, Git, CI/CD, Cloud-Native Applications.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise backend and data-processing applications using **Python, Flask, REST APIs, SQL, and microservices**, supporting large-scale operational workflows and data-driven business processes across distributed systems.
- Built reusable Python components for **data transformation, workflow automation, API integration, business logic, scheduled processing, configuration management, and enterprise service communication**.
- Designed data-processing workflows using **Python, SQL, PostgreSQL, MySQL, batch pipelines, data modeling, and relational datasets**, converting raw transactional information into reliable outputs for business applications and reporting.
- Applied **data-quality controls, reconciliation logic, validation rules, exception handling, logging, and monitoring** to improve consistency and reliability across business-critical data flows and integrated systems.
- Integrated enterprise platforms using **REST APIs, JSON, authentication, authorization, event-driven architecture concepts, and CI/CD workflows**, supporting secure communication across backend services and downstream applications.
- Collaborated with architects, business analysts, engineers, and client stakeholders to **break down complex requirements, interpret business data, identify implementation options, and deliver actionable technical solutions** aligned with organizational objectives.
- Created reusable **service modules, API patterns, technical documentation, implementation guides, and knowledge-sharing materials**, supporting junior team members and reinforcing professional and technical engineering standards.

Environments: Python, Flask, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Data Modeling, Data Pipelines, Batch Processing, Data Quality Controls, Event-Driven Architecture, Git, CI/CD, Linux, Cloud Concepts, Logging, Monitoring, Enterprise Integration.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.